

Ucchwas Talukder Utsha

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[GitHub](#) | [Google Scholar](#) | [LinkedIn](#) | U.S. Permanent Resident · No sponsorship required

SUMMARY

Computer scientist and Stanford postdoctoral scholar creating **LLM reasoning, graph-learning, and biomedical AI systems**. Research spans cellular foundation models, reasoning over complex networks, biosignal decoding, and deployable edge inference, with experience carrying ideas from experimental design through GPU training, rigorous evaluation, APIs, and mobile deployment.

TECHNICAL SKILLS

ML / LLMs: PyTorch, TensorFlow, Hugging Face Transformers, ViTs, GNNs, graph Transformers, self-supervised learning, masked autoencoders, LoRA/adapters, RAG/GraphRAG, multi-agent systems, constrained decoding

Training / Systems: Mixed precision, gradient checkpointing, Slurm, vLLM, Google Cloud, Docker, TensorFlow Lite, experiment tracking, ablations, bootstrap confidence intervals

Programming / Software: Python, C/C++, Java, SQL, FastAPI, Spring Boot, REST APIs, Git, CI/CD

RESEARCH EXPERIENCE

Postdoctoral Scholar | Stanford University

Aug 2026 – Present

Radiation Physics, School of Medicine | Palo Alto, CA

- Developed masked-autoencoder ViT pretraining for proteomics foundation models, scaling to a 302M-parameter ViT-Large encoder trained on **69K real proteomes** from 1,067 studies and **230M synthetic** profiles; evaluated the frozen encoder across downstream tasks using cohorts excluded from pretraining.
- Designed a training-only synthetic-task generator that preserves protein correlations, study-specific missingness, and nonlinear target relationships; integrated classification and regression with real-data masked-autoencoder training.
- Created **BioGlyph**, a deterministic topology-to-language compiler that translates graph structure into interpretable roles, supporting evidence, and perturbation consequences for frozen LLMs; achieved **70.6%** system accuracy versus **48.7%** with edge-list prompts across the eight-network main benchmark.
- Established a graph-reasoning benchmark with **1,239 questions** across seven reasoning families, complemented by studies spanning **20 networks**; evaluated raw graph representations, GCN/ GraphSAGE/ GAT/ GIN-derived roles, and frozen LLM readers through controlled ablations and bootstrap confidence intervals.
- Orchestrated graph analysis, question-aware subgraph retrieval, and LLM reasoning into multi-agent biomedical workflows; served **4-bit Qwen3-8B** behind FastAPI/vLLM on Google Cloud for interactive inference.

Graduate Research Assistant | Texas Tech University

Aug 2022 – Jul 2026

Computer Science | Lubbock, TX

- Designed an end-to-end facial-**EMG silent-speech** stack using temporal convolutions, residual blocks, a four-layer Transformer, adapter/soft-prompt conditioning, and joint autoregressive/CTC training.
- Compared 10 decoding strategies and isolated the value of LLM conditioning through ablations; grammar-constrained CTC reduced WER from **29.5% to 23.5% (20.3% relative)** on 50 held-out utterances.
- Created a GAN-augmented CNN-LSTM cardiac classifier with **98.44%** MIT-BIH test accuracy and translated it into a **1.69 MB TensorFlow Lite** model running at **4.6 ms/beat** inside a BLE-connected Android application.
- Derived respiratory rate directly from ECG and reached **MAE $\leq$ 0.36 breaths/min** against a reference belt across four activities in five participants, enabling respiration sensing without a dedicated respiratory channel.
- Implemented an Apnea-ECG neural classifier as a mobile TensorFlow Lite model, reproducing **91.24-93.26%** validation accuracy at approximately **5 ms** per inference.

EDUCATION

PhD, Computer Science | Texas Tech University

2026

GPA: 4.0/4.0

BSc, Computer Science and Engineering | BUET

2021

Bangladesh University of Engineering and Technology

INDUSTRY EXPERIENCE

Software Engineer | Fiftytwo (Bording Group)

Aug 2021 – Jun 2022

Dhaka, Bangladesh and Glostrup, Denmark

- Delivered 52Retail POS features from requirements and technical design through release, improving checkout throughput by **approximately 15%**.
- Engineered C/C++ components and REST integrations connecting **POS** terminals, payment providers, and **ERP systems** across end-to-end retail transaction flows.
- Coordinated with product, engineering, and QA to scope releases and harden delivery through Bitbucket reviews, **CI/CD**, and **TestRail/Zephyr** regression coverage.

Software Engineer | Emerging IT Bangladesh Ltd.

Feb 2021 – Jul 2021

Dhaka, Bangladesh

- Implemented **Java/Spring Boot** services and **REST APIs** consumed by Android applications, taking backend features from endpoint design through client integration.
- Traced production failures across API and **Android** request flows, streamlined connectivity and endpoint behavior, and resolved defects affecting service-to-app responsiveness.

SELECTED PUBLICATIONS AND MANUSCRIPTS

- Language-encoded network topology enables large language models to reason about complex networks. *Submitted to Nature Methods, 2026.*
- Reducing Language-Prior Drift in EMG Silent Speech Recognition via NeuroSymbolic Constrained Decoding with Large Language Models. *In revision at IEEE Transactions on Audio, Speech and Language Processing, 2026.*
- LLM-Conditioned Silent Speech Recognition from Facial EMG Using NeuroSymbolic Decoding. *IEEE EMBC, 2026. Accepted.*
- Physiological Stress Monitoring with Chest-Worn ECG and Respiratory Sensing. *IEEE EMBC, 2026. Accepted.*
- GAN-enhanced hybrid deep learning model for real-time cardiac beat detection in the CardioHelp app with edge computing. *Smart Health 38, 100601, 2025.*

SELECTED PROJECTS

Graph to LLM / Graph-Native LGM | PyTorch, Graph Transformers, GNN baselines

- Designed a graph-native model for **raw graph-to-graph learning**. The model takes adjacency, edge values, and optional node features directly into a trainable graph Transformer, then reconstructs or translates the graph with a bilinear decoder. No text conversion or language tokenizer is required.
- Added relative random-walk probabilities (**RRWP**) as pairwise structural biases inside graph attention for TCGA normal-to-tumor graph prediction. Across **11 held-out cancers**, RRWP improved average precision for changed edges from **0.279 to 0.309** and AUROC from **0.703 to 0.731**. Compared the model with capacity-matched edge-aware GCNs and structural controls.

ReproPilot | Python, OpenAI API, SQLite

- Created an **AI research agent** that turns a paper PDF into a runnable reproducibility repository. It extracts the problem, data, model, metrics, and experimental setup, then generates training and evaluation code, configs, dependencies, tests, documentation, and a reproducibility report.
- Implemented schema-validated tool calls, PDF and file tools, restricted shell execution, and automatic test checks. Added SQLite event logging so prompts, tool calls, outputs, and execution decisions can be traced for each run.

MedGemma Navigator | Transformers, DICOMweb, FHIR R4, Gradio

- Built a **MedGemma-4B** clinical navigation prototype that retrieves medical images through DICOMweb and patient context through FHIR R4, then generates structured imaging summaries and pre-visit packets instead of simple image captions.

EMG Intent | PyTorch, Transformer, emg2pose

- Developed a model for **session-level action-intent decoding from 16-channel wrist sEMG**. Instead of predicting a single gesture, it learns program families, semantic action primitives, hand context, and exact **29-stage** activity identity from longer temporal context.
- Used pose only during training to provide **24 pose-grounded action tokens** and teacher representations. The EMG model learns the teacher's action structure through distillation and support-conditioned calibration, while inference remains **EMG-only**.